

Final Report Guideline for Advanced Artificial Intelligence Course (ChinaXiv-Style Survey Paper Format)

1. Overall Positioning & General Requirements

This guideline follows the standard specification of **ChinaXiv domestic academic survey papers** and fully aligns with the 2026 **Advanced Artificial Intelligence** graduate course syllabus of Harbin Institute of Technology, Shenzhen. The course covers eight core intelligent modules: Structural Intelligence, Visual Intelligence, Language Intelligence, Generative Intelligence, Graph Intelligence, Multimodal Intelligence, Reinforcement Intelligence, and Agent Intelligence.

The final project requires students to complete a **standard academic preprint survey paper or original research paper**, examining students' systematic mastery of eight major AI domains, ability to read frontier top-tier papers, analyze mainstream AI models, and summarize technical evolution and application rules. All reports must reflect:

- Systematic understanding and logical integration of the eight AI intelligence modules
- In-depth literature investigation, structured sorting, and comparative analysis of selected AI sub-directions
- Standard academic preprint writing paradigm, rigorous argumentation, and complete logical closed loop

Reference Format: ChinaXiv.org (Chinese Academy of Sciences Science Paper Preprint Platform) survey & preprint paper standard

Course Basic Information: Graduate-level Professional Elective | 32 Hours | 2 Credits | Theory + Project Integrated Teaching

Prerequisites: Proficient Python programming, basic mathematics (linear algebra, calculus, probability), fundamental machine learning knowledge

Course Assessment Basis: This final report accounts for **40% of the total course grade (Final Project)**

1.1 Front Matter Page Requirements

1.1.1 Title & Author Information

- **Paper Title:** Clearly define the survey/research theme, closely based on the eight course intelligence modules. Examples:

1. *Survey on Visual Intelligence: CNN Principles, Detection & Segmentation Technical Evolution*
2. *Investigation on LLM and Language Intelligence: Transformer Mechanism and Frontier Development*
3. *Technical Survey on Generative Intelligence: VAE, GAN and Diffusion Model Paradigms*
4. *Research Progress of Graph Intelligence and Reinforcement Intelligence in Modern AI Systems*

- **Author Information:** Student Name, Supervisor Name, School of Computer Science and Technology, Harbin Institute of Technology, Shenzhen, Advanced Artificial Intelligence Course Laboratory

1.1.2 Abstract (150–300 Words)

The abstract must be self-contained, covering **research background, objectives, methods, core results and conclusions**, closely tied to the eight AI module system of the course.

- **Background & Significance:** Briefly introduce the development context of modern advanced artificial intelligence, explain the positioning and interconnection of the eight core intelligent modules, and analyze the research gaps and practical demands of the selected direction.
- **Research Objectives:** Clarify the core purpose of the survey, such as sorting out technical evolution, comparing model performance, analyzing mechanism advantages and limitations, or summarizing application scenarios of a specific AI module.
- **Research Methods:** State literature sources (course handouts, NeurIPS/ICML/CVPR/ACL top-tier papers, open-source technical documents) and systematic investigation & comparative analysis methods.
- **Core Conclusions:** Summarize key findings, technical bottlenecks, application rules and future development trends of the investigated AI direction.

1.1.3 Keywords (3–5 Items)

Select terms from course core modules, e.g., Structural Intelligence, Visual Intelligence, Large Language Model, Generative AI, Graph Neural Network, Reinforcement Learning, Multimodal Fusion, LLM-based Agent

1.2 Core Body Chapter Structure

Chapter 1 Introduction

This chapter builds the overall research logic and highlights the connection with the Advanced AI course system.

- **1.1 Research Background and Significance:** Start from the overall framework of the course's eight intelligent modules, elaborate the development and iteration of modern advanced artificial intelligence technologies, explain the academic and industrial value of the selected research direction, and clarify the learning significance of integrating and deepening course knowledge.
- **1.2 Research Scope and Problem Definition:** Define clear research boundaries, focus on one or multiple correlated course modules (visual/language/generative/graph/reinforcement/multimodal/agent intelligence), and propose core research questions to be solved in this survey.
- **1.3 Domestic and Foreign Research Status:** Sort out the latest research progress of the target direction based on top-tier conference papers (NeurIPS, ICML, CVPR, ACL) specified in the course; summarize existing research strengths and deficiencies, and clarify the innovation and systematic value of this paper.
- **1.4 Report Organization:** Briefly overview the core content and logical arrangement of subsequent chapters.

Chapter 2 Theoretical Foundation: Eight Core AI Intelligence Modules (Course Knowledge Review)

This chapter systematically reviews the complete theoretical system of the course, lays a solid foundation for follow-up investigation and analysis, and reflects comprehensive mastery of course knowledge.

- **2.1 Structural Intelligence:** Review fundamental neural network theories, including neuron models, activation functions, MLP, CNN and RNN basic structures, as the underlying support of all advanced AI tasks.
- **2.2 Visual Intelligence:** Sort out CNN core principles, object detection, semantic segmentation and other classic computer vision task paradigms and technical characteristics.
- **2.3 Language Intelligence:** Elaborate sequence modeling theories, Transformer self-attention mechanism, and basic principles of large language models (LLMs).
- **2.4 Generative Intelligence:** Summarize the core principles and technical differences of classic generative models including VAE, GAN and Stable Diffusion.
- **2.5 Graph Intelligence:** Explain graph neural network theories, GCN, GAT and graph representation learning methods for non-Euclidean data modeling.
- **2.6 Multimodal Intelligence:** Review cross-modal fusion mechanisms and audio-visual integration technologies for multimodal AI tasks.
- **2.7 Reinforcement Intelligence:** Sort out basic RL algorithms, multi-agent system mechanisms, and core principles of DQN and PPO.

- **2.8 Agent Intelligence:** Introduce the design logic and workflow construction of LLM-based intelligent agents.

Chapter 3 In-Depth Investigation of the Target AI Module / Direction

Focus on the selected core module(s), conduct systematic literature sorting, technical combing and paradigm analysis, which is the core body of the survey paper.

- **3.1 Technical Connotation and Core Paradigm:** Define the technical scope of the investigated direction, clarify its positioning in the eight major AI modules, and analyze the core problem-solving logic and task scenarios.
- **3.2 Technical Evolution and Classic Schemes:** Sort out the technical iteration context from traditional methods to cutting-edge schemes, analyze the structural improvement and performance breakthrough of each generation of classic models, and cite top-tier conference papers for support.
- **3.3 Key Technical Details and Mechanism Analysis:** Elaborate core algorithm principles, model structure design, mathematical logic and implementation details; explain the internal mechanism of performance advantages based on course theories.
- **3.4 Current Technical Bottlenecks:** Summarize the core limitations of existing technologies in terms of accuracy, efficiency, generalization, deployment cost or scenario adaptability.

Chapter 4 Comparative Analysis and Frontier Research Progress

Carry out horizontal comparison of mainstream algorithms/models, and sort out the latest frontier research trends of the target direction.

- **4.1 Comparative Analysis of Mainstream Methods:** Compare different models/algorithms in terms of structural design, computational complexity, task performance, applicable scenarios and resource consumption, and summarize their adaptability rules.
- **4.2 Frontier Research Trends:** Based on recent NeurIPS/ICML/CVPR/ACL papers, summarize emerging technical directions, innovative improvement ideas and latest application breakthroughs.
- **4.3 Module Cross-Fusion Research:** Analyze the fusion application potential of the target module with other intelligent modules (e.g., multimodal fusion of visual and language intelligence, agent system construction based on LLMs and reinforcement learning).

Chapter 5 Discussion and Analysis

- **5.1 Core Research Findings:** Refine systematic conclusions from the full-text investigation, summarize the technical characteristics, applicable boundaries and

development laws of the target AI direction.

- **5.2 Response to Initial Research Questions:** Precisely answer all core questions proposed in the introduction based on the sorting and analysis results.
- **5.3 Course Knowledge Reflection and Expansion:** Combine the eight-module course system to reflect on the deepened understanding of course knowledge, and explain how frontier researches expand and innovate on the basis of basic course theories.
- **5.4 Research Limitations:** Objectively analyze the deficiencies of this survey in literature coverage, analysis depth and research perspective.

Chapter 6 Conclusion and Future Outlook

- **6.1 Main Conclusions:** Concisely summarize the full-text investigation results, form a complete logical closed loop with the research background and research objectives, and highlight the course learning gains and systematic cognition.
- **6.2 Future Development Outlook:** Prospect the future development trend of the target AI module, including algorithm optimization, cross-module fusion, engineering deployment and industrial landing; put forward feasible follow-up research directions.

1.3 Back Matter Requirements

References

Adopt **GB/T 7714 standard** (ChinaXiv mainstream format). Priority citation sources: course self-developed teaching materials, top-tier conference papers (NeurIPS, ICML, CVPR, ACL), classic AI monographs and open-source technical documents. All citations must be marked with in-text serial numbers.

Sample References:

- [1] Vaswani A, et al. Attention is all you need [C]//Proceedings of the 31st International Conference on Neural Information Processing Systems. 2017.
- [2] Goodfellow I, Pouget-Abadie J, Mirza M, et al. Generative adversarial networks [J]. Advances in Neural Information Processing Systems, 2014.
- [3] Course Teaching Group. Advanced Artificial Intelligence Course Comprehensive Teaching Materials [Z]. HITSZ, 2026.

2. Course Assessment Rules & Academic Specifications

2.1 Course Total Assessment Composition (100%)

Assessment Component	Weight	Detailed Description
Class Performance & Assignments	20%	Attendance, punctuality and active classroom discussion participation
Final Project (This Survey Paper)	40%	Complete qualified preprint survey/research paper based on eight AI modules and course experiments
Final Theoretical Exam	40%	Comprehensive assessment of core AI theories and classic case analysis capabilities

2.2 Format Specifications

- **Font:** Full text in Times New Roman; all level-1/level-2 chapter titles bolded
- **Line Spacing:** 1.5 times line spacing; page margin 2.5cm for all sides
- **Page Number:** Bottom center, starting from the main body
- **Figures & Tables:** Uniform numbering, clear titles and source annotations, with mandatory in-text citation and textual explanation

2.3 Academic Norms & Word Count

- Strictly abide by academic integrity; all quoted content must be marked with citations; no plagiarism or falsified conclusions
- Unified terminology consistent with course teaching materials and frontier paper expressions
- **Word Count:** 5000–10000 words for the main body (excluding abstract, references and appendix)

3. Recommended Survey Topics (Fully Adapted to Eight AI Modules)

All topics strictly match the 2026 Advanced Artificial Intelligence course system, supporting standard preprint survey paper writing:

1. **Structural Intelligence Survey:** Research on Structural Evolution and Performance Comparison of MLP, CNN and RNN Basic Neural Networks
2. **Visual Intelligence Survey:** Technical Evolution and Application Analysis of

CNN-based Object Detection and Semantic Segmentation

3. **Language Intelligence Survey:** Transformer Mechanism Analysis and Frontier Development of Large Language Models
4. **Generative Intelligence Survey:** Comparative Study of VAE, GAN and Diffusion Generative Model Paradigms
5. **Graph Intelligence Survey:** Principle Analysis and Application Progress of GCN and GAT Graph Neural Networks
6. **Multimodal Intelligence Survey:** Research Progress of Cross-Modal Fusion and Audio-Visual Integration Technologies
7. **Reinforcement Intelligence Survey:** Algorithm Comparison of DQN/PPO and Frontier Research of Multi-Agent Reinforcement Learning
8. **Agent Intelligence Survey:** Workflow Design and Application Research of LLM-Based Intelligent Agents
9. **Cross-Module Comparative Survey:** Performance Adaptability Analysis of Different AI Intelligent Modules in Typical Downstream Tasks

4. Course Expected Learning Outcomes (Paper Evaluation Basis)

The completed report must fully reflect the following course learning capabilities:

- Systematic mastery of core theories covering eight major AI intelligent domains
- Capability to read, understand and summarize top-tier frontier AI papers
- Ability to independently sort out technical systems and complete standardized academic survey writing
- Independent learning and frontier AI technology tracking capabilities